Eg α and β are root

$f(x) \Rightarrow 2x$	$f(x) \Rightarrow x+5$	$f(x) \Rightarrow \frac{2x-1}{x}$
$f(\alpha) \Rightarrow 2\alpha$	$\alpha+5$	$\frac{2\alpha-1}{\alpha}$
$f(\beta) \Rightarrow 2\beta$	$\beta+5$	$\frac{2\beta-1}{\beta}$

Ex) Let α and β, be the roots of the Quadratic Eq^n $ax^2+bx+c \Rightarrow 0$, $c \neq 0$. find the Quadratic eq^n whose roots are $\frac{1-\alpha}{\alpha}$ and $\frac{1-\beta}{\beta}$

M-1 M-2

$\Rightarrow$ α and β

$ax^2+bx+c \Rightarrow 0$

$c \neq 0$, {C $\Rightarrow$ Constant}

$\rightarrow a\alpha^2+b\alpha+c \Rightarrow 0$

$x \Rightarrow \frac{1-\alpha}{\alpha} \Rightarrow \left(\frac{1}{\alpha}-1\right)$

$\rightarrow a\left(\frac{1-\alpha}{\alpha^2}\right)^2+b\alpha^2+c \Rightarrow 0$

$\rightarrow \alpha^2 a(1-\alpha)^2+b\alpha^2+c\alpha^2 \Rightarrow 0$

$\rightarrow \alpha^2 a(1-$

Quadratic Eq^n formula

$x^2-(\text{Sum of roots})x+(\text{product of roots}) \Rightarrow 0$

$x^2-(\alpha+\beta)x+(\alpha\beta) \Rightarrow 0$

Sum of Roots $\Rightarrow \frac{1-\alpha}{\alpha}+\frac{1-\beta}{\beta}$

$\frac{-b}{a} \Rightarrow \frac{\beta-\alpha\beta+\alpha-\alpha\beta}{\alpha\beta}$

$\frac{-b}{a} \Rightarrow \frac{\alpha+\beta-2\alpha\beta}{\alpha\beta}$

$\frac{-b}{a} \Rightarrow \frac{\alpha+\beta}{\alpha\beta}-2$

$\alpha\beta\left(\frac{-b}{a}+2\right) \Rightarrow \alpha+\beta$

$\rightarrow$ Product $\Rightarrow \frac{1-\alpha}{\alpha} \times \frac{1-\beta}{\beta} \Rightarrow \frac{c}{a}$

$\rightarrow \frac{1-\beta-\alpha+\alpha\beta}{\alpha\beta}$

$\rightarrow 1-1(\alpha+\beta)+\alpha\beta$

Eg α and β are root

$f(x) \Rightarrow 2x$	$f(x) \Rightarrow x+5$	$f(x) \Rightarrow \frac{2x-1}{x}$
$f(\alpha) \Rightarrow 2\alpha$	$\alpha+5$	$\frac{2\alpha-1}{\alpha}$
$f(\beta) \Rightarrow 2\beta$	$\beta+5$	$\frac{2\beta-1}{\beta}$

Ex) Let α and β, be the roots of the Quadratic Eq^n $ax^2+bx+c \Rightarrow 0$, $c \neq 0$. find the Quadratic eq^n whose roots are $\frac{1-\alpha}{\alpha}$ and $\frac{1-\beta}{\beta}$

M-1 M-2

$\Rightarrow \alpha$ and β

$ax^2+bx+c \Rightarrow 0$

$c \neq 0$, {C $\Rightarrow$ Constant}

$\rightarrow a\alpha^2+b\alpha+c \Rightarrow 0$

$x \Rightarrow \frac{1-\alpha}{\alpha} \Rightarrow \left(\frac{1}{\alpha}-1\right)$

$\rightarrow a\left(\frac{1-\alpha}{\alpha^2}\right)^2 + b\alpha^2 + c \Rightarrow 0$

$\rightarrow \alpha^2 a(1-\alpha)^2 + b\alpha^2 + c\alpha^2 \Rightarrow 0$

$\rightarrow \alpha^2 a(1-$

Quadratic Eq^n formula

$x^2 - (\text{Sum of roots})x + (\text{product of roots}) \Rightarrow 0$

$x^2 - (\alpha+\beta)x + (\alpha\beta) \Rightarrow 0$

Sum of Roots $\Rightarrow \frac{1-\alpha}{\alpha} + \frac{1-\beta}{\beta}$

$\frac{-b}{a} \Rightarrow \frac{\beta-\alpha\beta+\alpha-\alpha\beta}{\alpha\beta}$

$\frac{-b}{a} \Rightarrow \frac{\alpha+\beta-2\alpha\beta}{\alpha\beta}$

$\frac{-b}{a} \Rightarrow \frac{\alpha+\beta}{\alpha\beta} - 2$

$\alpha\beta\left(\frac{-b}{a}+2\right) \Rightarrow \alpha+\beta$

$\rightarrow$ Product $\Rightarrow \frac{1-\alpha}{\alpha} \times \frac{1-\beta}{\beta} \Rightarrow \frac{c}{a}$

$\rightarrow \frac{1-\beta-\alpha+\alpha\beta}{\alpha\beta}$

$\rightarrow \frac{1-1(\alpha+\beta)+\alpha\beta}{\alpha\beta}$

⇒ Both are Symmetric Roots

α, β are roots of $ax^2+bx+c \Rightarrow 0$

$a\alpha^2+b\alpha+c \Rightarrow 0$

→ $x \Rightarrow \frac{1-\alpha}{\alpha}$

$\alpha \Rightarrow \frac{1}{x+1}$

$a\left[\frac{1}{x+1}\right]^2 + b\left[\frac{1}{x+1}\right] + c \Rightarrow 0$

$\frac{a}{(x+1)^2} + \frac{b}{(x+1)} + c \Rightarrow 0$

→ $a+b(x+1)+c(x+1)^2 \Rightarrow 0$

⇒ ~~Mul Reciprocal~~

⇒ $cx^2+(b+2c)x+(a+b+c) \Rightarrow 0$

M-2

⇒ Sum of Roots ⇒ $\alpha+\beta \Rightarrow \frac{-b}{a}$

$\frac{1-\alpha}{\alpha} + \frac{1-\beta}{\beta} \Rightarrow \frac{-b}{a}$

$\frac{1}{\alpha} - 1 + \frac{1}{\beta} - 1 \Rightarrow \frac{-b}{a}$

$\frac{1}{\alpha} + \frac{1}{\beta} \Rightarrow -\frac{b}{a} + 2 \Rightarrow \frac{-b+2a}{a}$

{ The value $\alpha+\beta$, $\alpha\beta$ always comes from first Quadratic }

$\Rightarrow \frac{\alpha+\beta}{\alpha\beta} \Rightarrow \frac{-b+2a}{a}$

$\alpha+\beta \Rightarrow \frac{-b+2a}{a}[\alpha\beta]$

$\{\alpha\beta \Rightarrow \frac{c}{a}\}$

$\alpha+\beta \Rightarrow \frac{(-b+2a)c}{a^2}$ $\Rightarrow$ Sum of Roots

→ Product of roots $\Rightarrow$ ~~$\frac{c}{a}$~~

$\left(\frac{1}{\alpha}-1\right)\left(\frac{1}{\beta}-1\right) \Rightarrow \frac{c}{a}$

$\frac{1}{\alpha\beta} - \frac{1}{\alpha} - \frac{1}{\beta} - 1 \Rightarrow \frac{c}{a}$

$\frac{1}{\alpha\beta} - \left[\frac{\alpha+\beta}{\alpha\beta}\right] - 1 \Rightarrow \frac{c}{a}$

$\frac{1-[\alpha+\beta]-\alpha\beta}{\alpha\beta} \Rightarrow \frac{c}{a}$

$\alpha\beta \Rightarrow \left[1-\left[\frac{-b}{a}\right]-\left[\frac{c}{a}\right]\right] a$

c

$\alpha\beta \rightarrow \frac{a+b+c}{c}$

{ Key Mistake }

→ Quadratic Eq^n $\Rightarrow$

$x^2 - (\text{Sum of roots})x + c \Rightarrow 0$

$x^2 - \left\{\frac{-bc+2ac}{a^2}\right\} + \frac{a+b+c}{c} \Rightarrow 0$

→ $x^2 + \left\{\frac{bc-2ac}{a^2}\right\}x + \frac{a+b+c}{c} \Rightarrow 0$

$x^2 + \left(\frac{bc^2-2ac^2}{a^2c}\right)x + a+b+c \Rightarrow 0$

→ $a^2x^2c + cx(b-2ac) + a+b+c \Rightarrow 0$

Ex) If (α, β) are the roots of the Quadratic Eq^n $ax^2+bx+c \Rightarrow 0$, then find the Quadratic Eq whose roots are $\left\{\frac{\alpha}{\beta}, \frac{\beta}{\alpha}\right\}$

⇒ Lets apply Swap test to see if Roots are Symmetrical.

The Swap test is:

1) Wherever You see ~~β~~ α, replace it with β
2) Wherever You see β, replace it with α.
3) See if the pair of roots is the same (possibility in different order)

Remember: The order of Roots dosent Mater

E.g

$2\alpha, 2\beta$ → Swap, $\alpha \to \beta$, $\beta \to \alpha$ → $2\beta, 2\alpha$

$\frac{\alpha}{\beta}, \frac{\beta}{\alpha}$ → $\alpha \to \beta$, $\beta \to \alpha$ → $\frac{\beta}{\alpha}, \frac{\alpha}{\beta}$

order dosnt mater and Yes they are Symmetrical



→ "α" and "β" are the roots

⇒ $ax^2+bx+c \Rightarrow 0$

→ $a\alpha^2+b\alpha+c \Rightarrow 0$ — (1)

Cause when u solve it the β is coming which it should't

→ $\alpha \Rightarrow \beta$

Substitue in (1)

$a(x\beta)^2+b(x\beta)+c \Rightarrow 0$

$ax^2\beta^2+bx\beta+c \Rightarrow 0$

→ multiply and divide.

$\left[\frac{\alpha^2}{\beta\alpha}, \frac{\beta^2}{\alpha\beta}\right]$ #

→ $\beta\alpha \Rightarrow \frac{c}{a}$

$x \Rightarrow \frac{\alpha 2}{c/a}$

$\rightarrow a\left[\left[\frac{a\alpha^2}{c}\right]\right]^2+b\left[\frac{\alpha 2}{c}a\right]+c \Rightarrow 0$

=

$\Rightarrow \frac{a^3\alpha^4}{c}+\frac{ba\alpha^2}{c}+c \Rightarrow 0$

$\Rightarrow a^3\alpha^4+ba\alpha^2+c^2 \Rightarrow 0$

$\Rightarrow x \Rightarrow \frac{\alpha 2}{c/a} \Rightarrow \sqrt{\frac{cx}{a}} \Rightarrow \alpha$

$\Rightarrow a\left[\frac{\sqrt{cx}}{\sqrt{a}}\right]^2+b\left[\frac{\sqrt{cx}}{\sqrt{a}}\right]+c \Rightarrow 0$

M-2 Sum of Roots $\Rightarrow \frac{-b}{a}$

$$\frac{\alpha}{\beta} + \frac{\beta}{\alpha} \Rightarrow \frac{-b}{a}$$

$$\frac{\alpha^2+\beta^2}{\alpha\beta} \Rightarrow \frac{-b}{a}$$

$$\frac{(\alpha+\beta)^2 - 2\alpha\beta}{\alpha\beta}$$

$$\frac{(\alpha+\beta)^2}{\alpha\beta} \Rightarrow \frac{-b}{a} + 2$$

$\left\{ \alpha\beta \Rightarrow \frac{c}{a} \right\}$

~~$(\alpha+\beta)^2 \Rightarrow (-b)^2 - 2(\alpha\beta)$~~

$$\rightarrow \frac{\left(\frac{-b}{a}\right)^2 - 2\left(\frac{c}{a}\right)}{\left(\frac{c}{a}\right)}$$

$$\rightarrow \left\{\frac{b^2}{a^2} - \frac{2}{a}\right\} a \Rightarrow \left\{\frac{b^2 - 2ac}{a^2}\right\} a$$

$$\underline{\text{Sum of roots}} \Rightarrow \frac{b^2 - 2ac}{ac}$$

- Product of Roots $\Rightarrow \frac{\alpha}{\beta} \cdot \frac{\beta}{\alpha}$

$$\Rightarrow \frac{\alpha\beta}{\alpha\beta} \Rightarrow \frac{c}{c} \Rightarrow 1$$

- Quadratic Eq^n $\Rightarrow x^2 - [\text{Sum of roots}]x + \text{product of roots}$

→ $x^2 - \left(\frac{b^2-2ac}{ac}\right)x + 1 \Rightarrow 0$

→ $acx^2 - (b^2+2ac)x + ca \Rightarrow 0$

→ $acx^2 - (b^2+2ac)x + ca \Rightarrow 0$

Ex) → If $\alpha \neq \beta$ but $\alpha^2 = 5\alpha - 3$, $\beta^2 \Rightarrow 5\beta - 3$, $\alpha^2 - 5\alpha + 3 \Rightarrow 0$, $\beta^2 - 5\beta + 3 \Rightarrow 0$

Then the Eq^n whose roots are $\frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$ is

→

$\alpha \& \beta \neq$ not same

a) $\alpha^2 \Rightarrow 5\alpha - 3$ b) $\beta^2 + 3 \Rightarrow 5\beta$

$\frac{\alpha^2+3}{5} \Rightarrow \alpha$ $\beta \Rightarrow \frac{\beta^2+3}{5}$

→ Sum of roots $\Rightarrow \alpha + \beta$

$\Rightarrow \frac{\alpha^2+3+\beta^2+3}{5} \Rightarrow \frac{\alpha^2+\beta^2+6}{5}$

$\alpha+\beta \Rightarrow \frac{(\alpha+\beta)^2 - 2\alpha\beta + 6}{5}$

$\rightarrow \frac{\left(\frac{-b}{a}\right)^2 - 2\left(\frac{c}{a}\right) + 6}{5} \Rightarrow \frac{\frac{b^2}{a^2} - \frac{2c}{a} + 6}{5}$

$\rightarrow \frac{b^2 - 2ac + 6a^2}{5a^2}$

a → $\alpha^2 - 5\alpha + 3 \Rightarrow 0$

$(x)^2 - 5(x) + 3 \Rightarrow 0$

{Think}

$\Rightarrow$ Sum of roots $\Rightarrow \frac{-(-5)}{1} \Rightarrow 5$

Product $\Rightarrow \frac{3}{1} \Rightarrow 3$

b) $\beta^2 - 5\beta + 3 \Rightarrow 0$

$(x)^2 - 5(x) + 3 \Rightarrow 0$

→ coefficient = 1

$ax^2+bx+c=0$

→ $x^2+5x+3=0$ — (1)

→ $1(x)^2-5(x)+3=0$

⇒ $x \Rightarrow \frac{\alpha}{\beta}$ → multiply and divide by α

$\Rightarrow \frac{\alpha^2}{\beta\alpha}$

$\Rightarrow \frac{\alpha^2}{ca} \rightarrow 3$

$3x \Rightarrow \alpha^2$

$\alpha \Rightarrow \sqrt{3x}$ — (2)

→ put (2) in (1)

→ $(\sqrt{3x})^2-5(\sqrt{3x})+3=0$

→ $3x-5\sqrt{3x}+3=0$

Ans-2

x^2- (Sum of Roots)$x+$ (Product of roots) $=0$

→ Sum of Roots $= \frac{\alpha}{\beta}+\frac{\beta}{\alpha}$

$\Rightarrow \frac{\alpha^2+\beta^2}{\alpha\beta} \Rightarrow \frac{(\alpha+\beta)^2-2\alpha\beta}{\alpha\beta}$

$\{\alpha\beta=3,\ \alpha+\beta=5\}$

$\Rightarrow \frac{(5)^2-2(3)}{3} \Rightarrow \frac{25-6}{3} \Rightarrow \frac{19}{3}$

→ Product $\Rightarrow \frac{\alpha}{\beta}\cdot\frac{\beta}{\alpha} \Rightarrow \frac{3}{3} \Rightarrow 1$

→ $x^2 - (\text{Sum of roots})x + (\text{product of roots}) \Rightarrow 0$

→ $x^2 - \left(\frac{19}{3}\right)x + 1 \Rightarrow 0$

→ $3x^2 - 19x + 3 \Rightarrow 0$

Ex) The Quadratic Eq^n $x^2+mx+n \Rightarrow 0$ has roots which are twice those of $x^2+px+m \Rightarrow 0$ and m, n and $p \neq 0$. find the value of $\frac{n}{p}$.

⇒ α, β (1) $\quad$ $2\alpha, 2\beta$ (2)

$x^2+mx+n \Rightarrow 0$

$(\alpha)^2+m(\alpha)+n \Rightarrow 0$

$(\beta)^2+m(\beta)+n \Rightarrow 0$

Sum of Roots $\Rightarrow -\frac{b}{a}$

$(\alpha+\beta) \Rightarrow \frac{-m}{1}$

$\Rightarrow -m$

product $\Rightarrow \frac{n}{1} \Rightarrow n \Rightarrow \alpha.\beta$

$x^2+px+m \Rightarrow 0$

Sum of Roots $\Rightarrow -\frac{p}{1}$

$2\alpha+2\beta \Rightarrow (-p)$

$2(\alpha+\beta) \Rightarrow \Rightarrow -2p$

$\Rightarrow -p/2$

product $\Rightarrow 4(\alpha.\beta) \Rightarrow 4(m)$

$\Rightarrow x^2 - ($

→ ~~$\frac{n}{\beta} \Rightarrow \frac{\alpha.\beta}{\alpha+\beta} \Rightarrow \frac{4m}{-m} \Rightarrow -4$~~

→ (1)

⇒ $x^2 - (\text{Sum of Roots})x + (\text{product of root}) \Rightarrow 0$

$x^2+mx+n \Rightarrow 0$

2) $x^2 - (-2p)x + 4m \Rightarrow 0$

$x^2+2px+4m \Rightarrow 0$

Sum of roots $\Rightarrow -\frac{2p}{1} \Rightarrow (-2p)$ || product $\Rightarrow 4m$

1) $x^2 + mx + n \Rightarrow 0$.
2) $x^2 + 2px + 4m \Rightarrow 0$

→ Comparing

a)

$m \Rightarrow 2p$

$n \Rightarrow 4m$
$n \Rightarrow 4(2p)$
$\frac{n}{p} \Rightarrow 8$

Ex) If α, β are roots of the eqn, $x^2 - 2x + 3 \Rightarrow 0$, find the Eqn whose roots are $\alpha^3 - 3\alpha^2 + 5\alpha - 2$ and $\beta^3 - \beta^2 + \beta + 5$.

$x^2 - 2x + 3 \Rightarrow 0$
$x^2 - 3x + x + 3 \Rightarrow 0$
$x(x-3) + (x-3) \Rightarrow 0$
$(x-3)(x-1) \Rightarrow 0$
$x \Rightarrow 3$ and 1.

$x^2 + x - 3x + 3 \Rightarrow 0$
$x(x+1) - 3(x+1)$

→ Sum of roots $\Rightarrow \frac{-b}{a}$

$\Rightarrow \frac{-(-2)}{4} \Rightarrow \frac{2}{4} \Rightarrow \alpha + \beta$

→ Product of Roots $\Rightarrow \alpha \cdot \beta \Rightarrow \frac{c}{a} \Rightarrow 3 \Rightarrow \alpha \cdot \beta$

Every time You see higher Power $\alpha^3 \cdot \alpha^4 \alpha^5$ you come deduce it by using Quadratic Eq^n

→ $x^2 - 2x + 3 \Rightarrow 0$

$\alpha^2 - 2\alpha + 3 \Rightarrow 0$

$\alpha^2 \Rightarrow 2\alpha - 3$

a) → $\alpha^3 - 3\alpha^2 + 5\alpha - 2$

$\alpha(\alpha^2) \Rightarrow \alpha(2\alpha - 3)$

$\Rightarrow 2\alpha^2 - 3\alpha$

$\Rightarrow 2(2\alpha - 3) - 3\alpha$

$\Rightarrow 4\alpha - 6 - 3\alpha$ ↷ $-3(2\alpha - 3)$

$\Rightarrow 4\alpha - 6 - 3\alpha - 3\alpha^2 + 5\alpha - 2 \Rightarrow \underline{1}$

β) $\beta^3 - 3\beta^2 + \beta + 5$

$\beta^2 \Rightarrow 2\beta - 3$

$\Rightarrow \beta(\beta^2) - \beta^2 + \beta + 5$

$\beta(2\beta - 3) - (2\beta - 3) + \beta + 5$

→ $2\beta^2 - 3\beta - 2\beta + 3 + \beta + 5$ ↷ ^{8}

→ $2(2\beta - 3) - 3\beta - 2\beta + 3 + \beta + 5$

→ $4\beta - 6 - 3\beta - 2\beta + \beta + 5$

→ $1\beta - 2\beta + 1\beta + 5 + 3 - 6$

→ 2

→ 2 and 1 are roots

∴ ⇒ Sum of Roots ⇒ $2 + 1 \Rightarrow 3$

∴ ⇒ Product of Roots ⇒ $2 \times 1 \Rightarrow 2$

Quadratic ⇒

$x^2 - (3)x + 2 \Rightarrow 0$

$x^2 - 3x + 2 \Rightarrow 0$

Theory of Equations [Relation b/w Roots and Coefficients]

1) Quadratic Equation $[ax^2 - bx + c \Rightarrow 0]$

$\alpha + \beta \Rightarrow \frac{-b}{a}$

$\alpha \cdot \beta \Rightarrow \frac{c}{a}$

α, β → Root

$-b$, a, c → Coefficient

Relation

Relation

Ex)

2. Cubic Equation

$\Rightarrow$ Eq: $ax^3 + bx^2 + cx + d \Rightarrow 0$

It has three Roots: α, β, γ

$\Rightarrow ax^3 + bx^2 + cx + d \Leftrightarrow a(x-\alpha)(x-\beta)(x-\gamma)$

multiply and divide by a

$\rightarrow x^3 + \left[\frac{b}{a}\right]x^2 + \left[\frac{c}{a}\right]x + \frac{d}{a} \Leftrightarrow x^3 - (\alpha+\beta+\gamma)x^2 + [\alpha\beta+\beta\gamma+\gamma\alpha]x - \alpha\beta\gamma$

$\therefore \alpha + \beta + \gamma \Rightarrow \frac{-b}{a}$

$\therefore \alpha\beta+\beta\gamma+\gamma\alpha \Rightarrow \frac{c}{a}$ {Sum of Roots taken 2 at a time}

$\therefore \alpha\beta\gamma \Rightarrow \pm\frac{d}{a}$

no. of roots ⇒ relations

E.g no. of roots → 2 → Relations → 2

no. of roots → 3 → Relations → 3

Ex) If $2x^3+mx^2-13x+n \Rightarrow 0$ has roots 2 and 3, then the value of m and n are:

→ $2x^3+mx^2-13x+n \Rightarrow 0$

→ It has 3 roots but given only 2 assume the 3rd root is γ

→ {2, 3, γ} → roots

⇒ $2x^3+mx^2-13x+n \Rightarrow 0$

$\therefore$ Sum of roots ⇒ $\alpha+\beta+\gamma \Rightarrow \frac{-b}{a}$

→ $2+3+\gamma \Rightarrow \frac{-m}{2}$ —(1), $2+3+\left[\frac{-5}{2}\right] \Rightarrow \frac{-m}{2}$, $4+0-5 \Rightarrow -m$

$+m \Rightarrow 5, -5$

$\therefore$ Sum of two roots ⇒ $\alpha\beta+\beta\gamma+\gamma\alpha \Rightarrow \frac{c}{a}$

→ $2(3)+3\gamma+\gamma 2 \Rightarrow \frac{c}{a} \Rightarrow \frac{-13}{2}$

$6+5\gamma \Rightarrow \frac{-13}{2}$

$5\gamma \Rightarrow \frac{-13}{2}-6$

$\gamma \Rightarrow -\frac{5}{2}$ —(2)

$\therefore$ Product of roots ⇒ $\alpha\beta\gamma \Rightarrow -\frac{d}{a}$

$+\frac{5}{2} \times 2 \times 3 \Rightarrow +\frac{n}{2}$, $n \Rightarrow 30$

$\left\{\begin{array}{l} m \Rightarrow -5 \\ n \Rightarrow 30 \end{array}\right\}$

3. Equation of degree 'n'

→ If $\alpha_1, \alpha_2, \alpha_3, \ldots \alpha_n$ are the root of the Eq^n

$f(x) \Rightarrow a_0x^n + a_1x^{n-1} + a_2x^{n-2} + a_3x^{n-3} + \ldots + a_{n-1}x + a_n = 0$

where $a_0, a_1, \ldots a_n$ are all real and $a_0 \neq 0$ then,

→ $a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \ldots + a_n \Rightarrow$

$x^n - (\alpha_1 + \alpha_2 \ldots \alpha_n)x^{n-1} + \ldots$

Theory of Equation

$\sum_{\alpha \to 0}^{\alpha \to \infty} \alpha_1 \Rightarrow \frac{-b}{a}$

$\sum_{i=0}^{i=\infty} \alpha_i \beta_i \Rightarrow \frac{c}{a}$

$\sum_{i=0}^{i=\infty} \alpha_i \beta_i \gamma_i \Rightarrow -\frac{d}{a}$

$\Rightarrow \sum \alpha_1 \Rightarrow -\frac{a_1}{a_0}$

$\Rightarrow \sum \alpha_1\alpha_2 \Rightarrow +\frac{a_2}{a_0}$

$\Rightarrow \sum \alpha_1\alpha_2\alpha_3 \Rightarrow -\frac{a_3}{a_0}$

$\Rightarrow \alpha_1\alpha_2\alpha_3 \ldots \alpha_0 \Rightarrow (-1)^n \frac{a_n}{a_0}$

Ex) If α, β, γ are the roots of the cubic x^3+qx+r ⇒ & then find the eqⁿ whose roots are $\alpha+\beta, \beta+\gamma, \gamma+\alpha$.

→ $x^3+qx+r \Rightarrow 0$; (α, β, γ) { $x^3+(0)x^2+qx+r \Rightarrow 0$ }

∴ $\alpha+\beta+\gamma \Rightarrow \frac{-b}{a} \Rightarrow \frac{0}{a} \Rightarrow 0$ ——(1)

∴ $\alpha\beta+\beta\gamma+\gamma\alpha \Rightarrow \frac{c}{a} \Rightarrow \frac{q}{1} \Rightarrow q$

∴ $\alpha\beta\gamma \Rightarrow \frac{-r}{1}$ [$\frac{-d}{a}$]

$\Rightarrow -r$

→ The roots $\alpha+\beta, \beta+\gamma, \gamma+\alpha$ { not Symmetric }

$\alpha+\beta \Rightarrow -\gamma$ [from (1)] ↓ $-\gamma$

$\beta+\gamma \Rightarrow -\alpha$ ↓ $-\alpha$

$\gamma+\alpha \Rightarrow -\beta$ ↓ $-\beta$

The roots are $-\gamma, -\alpha, -\beta$

→ $x^3+qx+r \Rightarrow 0$

↓ $x = -\alpha$ root

$\alpha^3+q\alpha+r \Rightarrow 0$

$-\alpha \Rightarrow x$

$\alpha \Rightarrow -x$

→ $(-x)^3 - q(x) + r \Rightarrow 0$

$-x^3 - qx + r \Rightarrow 0$

∴ $x^3 + qx - r \Rightarrow 0$

Ex) If α, β, γ are the roots of the Cubic $x^3+qx+r=0$ then find the eqⁿ whose roots are $\alpha\beta, \beta\gamma, \gamma\alpha$

→ Given.

$\Rightarrow x^3+qx+r=0$

$\Rightarrow x^3+0(x)^2+qx+r=0$

$\alpha+\beta+\gamma \Rightarrow 0$

$\alpha\beta+\beta\gamma+\gamma\alpha \Rightarrow q$

$\alpha\beta\gamma \Rightarrow -r$

$\Rightarrow \beta\gamma \Rightarrow \frac{r}{\alpha}$, $\gamma\alpha \Rightarrow \frac{r}{\beta}$, & $\beta\alpha \Rightarrow \frac{r}{\gamma}$

New	Roots
$\frac{-r}{\alpha}$	α
$\frac{-r}{\beta}$	β
$\frac{-r}{\gamma}$	γ

$\Rightarrow x^3+qx+r=0$

→ $\alpha^3+q\alpha+r=0$ {α is one of the roots} —(1)

$\frac{-r}{\alpha} \Rightarrow x$

$\alpha \Rightarrow -\frac{r}{x}$ {Substitute in (1)}

→ $\left(\frac{-r}{x}\right)^3 + q\left[\frac{-r}{x}\right] + r = 0$

$\Rightarrow \frac{-r^3}{x^3} - \frac{qr}{x} + r = 0$

→ $-r^3 - qrx^2 + rx^3 \Rightarrow 0$

⇒

~~$qrx^3 + (qr)x^2 + r^3 = 0$~~

∴ → $\underline{x^3 - qx^2 - r^2 \Rightarrow 0}$

Ex) If $x^2 - 3x + 2$ is one of the factors of the Expression $x^4 - px^2 + q$

⇒ Eq:

$x^4 - px^2 + q \Rightarrow 0$ $\{x^4 + (0)x^3 - px^2 + (0)x + q \Rightarrow 0\}$

↑ It has 4 factor

→ $\alpha + \beta + \gamma + \delta \Rightarrow -(-p) \Rightarrow p$

→ $\alpha\beta + \beta\gamma + \gamma\delta + \beta\gamma\delta + \delta\alpha + \gamma\alpha \Rightarrow 0$

→ $\alpha\beta\gamma\delta \Rightarrow -q$

⇒ $x^2 - 3x + 2$ is one of the factor

$x(x-1) - 2(x-1)$

$\underline{(x-1)(x-2)}$

⇒ We have Used factor Theorem

$\left[\frac{x^4 - px^2 + q}{x^2 - 3x + 2}\right] \Rightarrow Q(x)$

$x^4 - px^2 + q \Rightarrow Q(x)(x^2 - 3x + 2)$

∴ The idea is that if We make the R.H.S zero, then the L.H.S has to be zero.

That means the value of "x" that I put in R.H.S Should also satisfy L.H.S as well.

→ If we put

$x \Rightarrow 1$

$(x^4 - px^2 + q) \Leftrightarrow (x^2 - 3x + 2)\,Q(x)$

$1 - p + q \Rightarrow 0 \quad -(1)$

$q + 1 \Rightarrow p$

$x \Rightarrow 2$

$(16 - 4p + q) \Rightarrow 0 \quad -(2)$

$16 + q \Rightarrow 4p$

(2) − (1)

$\Rightarrow 16 - 4p + q \Rightarrow 0$

$-1 + p - q \Rightarrow 0$

$-3p \Rightarrow 15$

$\therefore p \Rightarrow -5 \quad -(3)$

(3) in (2)

→ $1 + (-5) + q \Rightarrow p$

$q \Rightarrow -6$

$16 - 4(-$

$4p \Rightarrow 16 + q$

$-p \Rightarrow -1 + q$

$3p \Rightarrow 15$

$p \Rightarrow 5 \quad -(3)$

(3) in (1)

$1 - 5 + q \Rightarrow 0$

$q \Rightarrow 4$

$\{\therefore p \to 5,\ q \Rightarrow 4\}$

Nature of Roots

→ The roots of the Quadratic Eqⁿ $ax^2+bx+c \Rightarrow 0$ is given by

$$x \Rightarrow \frac{-b \pm \sqrt{b^2-4ac}}{2a}$$

The Expression $D \Rightarrow b^2-4ac$ is Called the Discriminant of the Quadratic Eqⁿ

$D \Rightarrow$
- $> 0 \Rightarrow$ Real & Distict
- $0 \Rightarrow$ Real and Equal
- < 0 Complex (or) Imaginary

Ex) If $x^2+2x+2 \Rightarrow 0$

→ Let's find D first →

$b \Rightarrow 2,\ a \Rightarrow 1,\ c \Rightarrow 2$

$D \Rightarrow b^2-4ac$

$\Rightarrow (2)^2-4(1)(2)$

$\Rightarrow 4-8$

$\Rightarrow -4 < 0$ {Complex and imaginary} {$\sqrt{-1} \Rightarrow i$}

→ $x \Rightarrow \frac{-b \pm \sqrt{b^2-4ac}}{2a} \rightarrow \frac{-2 \pm \sqrt{-4}}{2} \rightarrow \frac{-2 \pm 2\sqrt{i}}{2}$

$x \Rightarrow \frac{-2+2\sqrt{i}}{2}$ and $\frac{-2-2\sqrt{i}}{2}$